

SIMATSENGINEERING(SAVEETHASCHOOLOFENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Traversal Challenge

Course Code:	CSA0309	Course Title:	Data Structures
CO Assessed:	CO3–Precision (BL3, BL5)	Assessment:	Assessment Tool 2 – Traversal Challenge
Weightage:	30% of CIA	Total Marks:	100
CO3 Statement:	Solve problems for optimized data storage and retrieval using Binary Search Trees, AVL Trees, B-Trees, Red-Black Trees, and Splay Trees.		
Student Name:	K. Nilavan	Reg. No.:	192511028
Section/Batch:	2 ND Year CSE	Date:	31/07/2026

Code of Conduct

I K. Nilavan (192511028) (Name/RegNo) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: K. Nilavan

Instructions

- This assessment tool contains 5 Traversal Challenge questions, Total: 100 Marks.
- Students can use any online/offline Traversal Challenge. Demonstrate tree.
- When a student answers using a simulator, in evaluation the answers should have captured screenshots after every major operation.
- Submit the report in PDF format with properly labelled screenshots as proof of completion.

Section/Topic		Focus	QNos	Marks
Tree Traversals		Traversal types, Inorder, Postorder and Preorder	5	100
GRAND TOTAL:			100 Marks	

Traversal Challenge

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Conceptual Understanding	20	Demonstrates thorough understanding of all core Data Structures concepts; answers accurately reflecting deep knowledge.	Shows good understanding with minor conceptual gaps.	Basic understanding; several misconceptions present.	Limited understanding; major concepts misunderstood.
Traversal Accuracy	20	Correctly completes all tree traversals (Preorder, Inorder, Postorder, Level Order) without errors.	Completes most traversals correctly with one minor error.	Correctly performs some traversals with multiple errors.	Unable to perform most traversals correctly.
Selection of Appropriate Traversal	30	Correctly selects the appropriate traversal for every given problem scenario.	Selects the correct traversal for most scenarios.	Selects appropriate traversal for only a few scenarios.	Frequently selects incorrect traversal methods.
Completion & Time Efficiency	20	Completes all challenges within the allotted time while maintaining accuracy.	Completes most challenges within the time limit.	Completes only part of the challenge or exceeds the time limit slightly.	Unable to complete the challenge within the allotted time.
Evidence Submission	10	Uploads complete screen shots showing successful completion, score, and student details.	Uploads screen shots with minor missing information.	Uploads incomplete or unclear screen shots.	No valid evidence submitted.

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

QUESTION-1

tree-graph-algorithm-visualizer.vercel.app/treetraversals

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Tree Traversals

Controls

Input
Input is level-by-level order.

30,20,39,10,25,35,42,15,23

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre In Post

Generate Default

Visualize

Start Simulation

```
graph TD; 30((30)) --> 20((20)); 30 --> 39((39)); 20 --> 10((10)); 20 --> 25((25)); 10 --> 15((15)); 10 --> 23((23)); 39 --> 35((35)); 39 --> 42((42));
```

tree-graph-algorithm-visualizer.vercel.app/treetraversals

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Tree Traversals

Controls

Input
Input is level-by-level order.

30,20,39,10,25,35,42,15,23

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre In Post

Generate Default

Visualize

Start Simulation

```
graph TD; 30((30)) --> 20((20)); 30 --> 39((39)); 20 --> 10((10)); 20 --> 25((25)); 10 --> 15((15)); 10 --> 23((23)); 39 --> 35((35)); 39 --> 42((42));
```

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Tree Traversals

Controls

Input
Input is level-by-level order.

30,20,39,10,25,35,42,15,23

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre In Post

Generate Default

Visualize

Start Simulation

```
graph TD; 30((30)) --> 20((20)); 30 --> 39((39)); 20 --> 10((10)); 20 --> 25((25)); 10 --> 15((15)); 10 --> 23((23)); 39 --> 35((35)); 39 --> 42((42));
```

QUESTION-2

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Tree Traversals

?

Input

Input is level-by-level order.

15,6,18,3,7,17,20,2,4,13,9

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

Visualize

Start Simulation

15 6 3 2 4 7 13 9 18 17 20

15

6 18

3 7 17 20

2 4 13 9

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Tree Traversals

?

Input

Input is level-by-level order.

15,6,18,3,7,17,20,2,4,13,9

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

Visualize

Start Simulation

2 3 4 6 13 7 9 15 17 18 20

15

6 18

3 7 17 20

2 4 13 9

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Tree Traversals

?

Input

Input is level-by-level order.

15,6,18,3,7,17,20,2,4,13,9

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

Visualize

Start Simulation

2 4 3 13 9 7 6 17 20 18 15

15

6 18

3 7 17 20

2 4 13 9

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QUESTION-3

HereweconsiderA,B,C,D,E,F,G,as1,2,3,4,5,6,7,

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Tree Traversals

?

Input

Input is level-by-level order.

1,2,3,4,5,6,7

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

Visualize

Start Simulation

4 2 5 1 6 3 7

1

2

3

4 5

6 7

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in

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Tree Traversals

?

Input

Input is level-by-level order.

1,2,3,4,5,6,7

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

Visualize

Start Simulation

1 2 4 5 3 6 7

1

2

3

4 5

6 7

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in

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Tree Traversals

?

Input

Input is level-by-level order.

1,2,3,4,5,6,7

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

Visualize

Start Simulation

4 5 2 6 7 3 1

1

2

3

4 5

6 7

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in

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QUESTION-5

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Input

Input is level-by-level order.

7,5,1,8,3,6,0,9,4,2

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

Visualize

Start Simulation

Tree Traversals

7 5 8 9 4 3 2 1 6 0

7

5

1

8

3

6

0

9

4

2

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Input

Input is level-by-level order.

7,5,1,8,3,6,0,9,4,2

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

Visualize

Start Simulation

Tree Traversals

9 8 4 5 2 3 7 6 1 0

7

5

1

8

3

6

0

9

4

2

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Input

Input is level-by-level order.

7,5,1,8,3,6,0,9,4,2

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

Visualize

Start Simulation

Tree Traversals

9 4 8 2 3 5 6 0 1 7

7

5

1

8

3

6

0

9

4

2

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QUESTION-4

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Input

Input is level-by-level order.

7,6,5,4,3,2,1

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

Visualize

Start Simulation

Tree Traversals

4

3

6

2

1

5

7

7

6

5

4

3

2

1

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Input

Input is level-by-level order.

7,6,5,4,3,2,1

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

Visualize

Start Simulation

Tree Traversals

7

6

4

3

5

2

1

7

6

5

4

3

2

1

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Input

Input is level-by-level order.

7,6,5,4,3,2,1

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

Visualize

Start Simulation

Tree Traversals

4

6

3

7

2

5

1

7

6

5

4

3

2

1

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